

Chapter32: The Renaissance of Indigeneity: Integrating IKS into NEP 2020

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Abstract

The National Education Policy (NEP) 2020 marks a paradigm shift in India's educational landscape, aiming to transform a legacy colonial structure into a vibrant, modern, and contextually rooted ecosystem (Ministry of Education, 2020). At the heart of this transformation is the integration of the Indian Knowledge System (IKS). Spanning mathematics, astronomy, metallurgy, ecology, philosophy, linguistics, and traditional medicine, IKS offers a holistic, eco-centric, and multidisciplinary framework for contemporary learning (AICTE, 2022). This review article evaluates the structural framework of integrating IKS into modern curricula, maps out the implementation successes achieved across primary and higher education, identifies pedagogical and operational challenges, and delineates future opportunities. It argues that while structural frameworks and institutional mandates have laid a robust foundation, overcoming epistemic biases, teacher training deficits, and linguistic barriers is critical to realizing the decolonial promise of NEP 2020.

Keywords: NEP 2020, Indian Knowledge System (IKS), Decolonial Education, Multidisciplinary Framework, Pedagogical Challenges

1. Introduction: The Decolonial Imperative in Indian Education

After Independence the Indian educational system worked largely within the structural and philosophical constraints inherited from British administration for a long time. This model of education came into existence through the Macaulay Minute of 1835 that advocated for the promotion of Western education and systematically marginalization of traditional knowledge and wisdom. The National Education Policy (NEP) 2020 seeks to rectify this asymmetry. Section 4.27 of the policy explicitly states: "The knowledge of India, including its rich tribal, classical, and modern traditions, will be seamlessly integrated into the curriculum at all levels of education, from foundational to higher stages, to instill pride, critical thinking, and a sense of national identity among learners" (Ministry of Education, 2020).

IKS is a pedagogical intervention/framework created by NEP to address current global crises. It addresses wide range of knowledge system that includes the Vedas, Upanishads, Nine types of philosophies (Astika, Nastika), Ayurveda, Jyotisha (astronomy), Mathematics. This comprehensive review examines the current state of IKS implementation five years post-policy announcement, analysing the underlying philosophy, integrations across educational tiers, successes, operational challenges, and highlights future opportunities to make India a world leader (Vishwa Guru).

2. Conceptual Framework of the Indian Knowledge System (IKS)

We must understand what constitutes IKS under the modern educational mandate. IKS basically represents a vast, empirical, and philosophical library accumulated over millennia across the Indian subcontinent. It is not a monolithic or purely religious construct of course.

2.1 The Dimensions of IKS

Ministry of Education's IKS Division covers three major domains (AICTE, 2022):

1. **Sciences and Mathematics:** Here comes the fundamental Sulba Sutras (foundational geometry), the concept of Shunya (zero) and decimal systems. It also encompasses the astronomical computations of Aryabhata and Bhaskara, plus the advanced metallurgical practices such as the production of wootz steel and rust-resistant alloys (e.g., the Iron Pillar of Delhi) (Raju, 2007).
2. **Humanities and Social Sciences:** It includes the classical linguistics (Vyakarana of Panini), epistemic validation methodologies (Pramana Shastra), ethics (Dharma and Niti), and complex socio-political governance models as outlined in the Arthashastra and Shanti Parva (Kapoor & Singh, 2021).
3. **Applied Arts and Lifestyles:** Sustainable agricultural practices (Krishi-Parashara), water management architectures, traditional medicine (Ayurveda, Unani, Siddha, Homeopathy), and the performing arts (Natya Shastra) fall in this domain.

2.2 Epistemic Pluralism and Multidisciplinary Learning

IKS operates on an assumption of interconnectedness, unlike Western Cartesian dualism, where mind differs from matter, the secular differs from the sacred, and humanity from nature. Here, knowledge is traditionally viewed as Vidya (experiential-transformative understanding) rather than just Avidya or Information (functional, material data).

This foundational paradigm is fed into the NEP 2020 by advocating for an absolute break from rigidity. By mandating a flexible choice of subjects—where a student can study Physics alongside Artificial Intelligence or Paninian grammar—the policy aligns modern pedagogy with the traditional Indian ideal of the Chatushashti Kalas (64 arts/sciences of comprehensive human development).

3. Structural Integration: From Foundational to Higher Education

The implementation of IKS requires systemic translation into institutional frameworks. This section breaks down how governing bodies have codified these mandates across different levels of instruction.

3.1 Foundational and Preparatory Stages (Ages 3–10)

Under the new 5+3+3+4 pedagogical structure, the National Curriculum Framework for Foundational Stage (NCF-FS) and School Education (NCF-SE) heavily incorporate IKS through playful, story-driven, and vernacular modalities (NCERT, 2023).

- **PanchakoshaAyama:** The framework for child development is explicitly modelled on the Taittiriya Upanishad's concept of the five sheaths of the human personality: Annamaya Kosha (physical), Pranamaya Kosha (vital/energy), Manomaya Kosha (mental/emotional), Vijnanamaya Kosha (intellectual), and Anandamaya Kosha (spiritual/aesthetic). Physical education, mindfulness, emotional regulation, and cognitive play are organized around these sheaths.
- **Storytelling and Folklore:** Traditional Indian fables from the Panchatantra, Hitopadesha, and regional tribal mythologies are utilized not merely for language acquisition but as tools for moral reasoning (Niti) and critical cognitive thinking.
- **Indigenous Games:** Local, traditional games like Kho-Kho, Kabaddi, Pallanguzhi, and Chaupar

are integrated into school routines to foster kinesthetic development and communal cohesion without requiring capital-intensive sports infrastructure.

3.2 Middle and Secondary Stages (Ages 11–18)

Here the curriculum emphasizes structural and historical analysis.

- **Textbooks and Cross-Curricular Infusion:** The National Council of Educational Research and Training (NCERT) has systematically revised textbooks to introduce dedicated modules on ancient Indian contributions to science, geography, and civic life (NCERT, 2023). For instance, mathematics modules now showcase the Baudhayana-Pythagoras theorem, while science textbooks highlight ancient rain-water harvesting structures.
- **Vocational Crafts:** Craft such as pottery, handloom weaving, organic farming, and traditional metallurgy are mandated in Secondary schools to provide exposure to local vocational crafts.

3.3 Higher Education and Research

Within the higher education sector under the guidelines of the University Grants Commission (UGC) and the All India Council for Technical Education (AICTE) there has been a sweeping integration of IKS.

- **Mandatory IKS Credit Courses:** The UGC issued guidelines requiring all undergraduate students, regardless of their major (B.Sc., B.A., B.Tech, B.Com), to complete a minimum of 5% of their total graduation credits in IKS-certified courses (UGC, 2023). This ensures that an engineering student is exposed to ancient Indian civil engineering, while a medical student learns the foundational principles of holistic health.
- **Establishment of the IKS Division:** Housed within the Ministry of Education at AICTE, the IKS Division acts as a central nodal agency funding interdisciplinary research centers, documentation projects, and curriculum development. Over 30 IKS Centers have been established in premier institutions like IITs, IIMs, and central universities to promote advanced research.

4. Implementation Successes: Empirical Milestone Marks

Five years into the gradual rollout of NEP 2020, several significant successes demonstrate the viability of this educational renaissance.

4.1 Institutionalization of Research and Funding Pathways

The Ministry of Education has funded hundreds of competitive research grants looking at ancient technologies through modern validation lenses. For example: Projects studying the therapeutic potential of Ayurveda formulations using advanced molecular biology, and evaluating ancient water-management systems in drought-prone areas using GIS mapping, are bridging traditional wisdom with contemporary empirical validation.

4.2 Proliferation of High-Quality Digital Learning Assets

To counter the acute shortage of physical textbooks, digital public learning infrastructure has yielded major dividends.

- **SWAYAM and MOOCs Platform:** The central government, in collaboration with premier IIT

faculty, launched comprehensive online courses titled "Introduction to Indian Knowledge Systems" on the SWAYAM portal. These courses have seen registrations from over a hundred thousand students and professionals, democratizing access to authentic scriptural and scientific translations.

- **IKS Repository:** A massive digital repository of ancient manuscripts, digitized texts, and instructional videos has been curated, providing access to global researchers.

4.3 Curriculum Customization by Technical Institutions

Elite technical institutes have pioneered the integration of IKS within core STEM paradigms.

- Institutes such as IIT Gandhinagar and IIT Indore have set up dedicated IKS cells, offering specialized programs on The History of Indian Science and Technology.
- **AICTE Model Curriculum:** AICTE successfully designed and rolled out a standard textbook on IKS for engineering students nationwide, standardizing the nomenclature, historical timelines, and conceptual frameworks (AICTE, 2022).

4.4 Revival of Local Craft and Artisanal Knowledge Systems

By linking Higher Educational Institutions (HEIs) with local clusters through programs like Unnat Bharat Abhiyan, universities have begun documenting and preserving localized tribal knowledge.

- In states like Madhya Pradesh and Odisha, universities have created credits for students who intern with tribal communities to document indigenous medicinal plants, agricultural seed-preservation techniques, and ecological conservation practices.

5. Systemic Challenges: The Roadblocks to Seamless Integration

Despite institutional momentum, the translation of IKS from a policy blueprint to classroom reality faces severe structural, intellectual, and logistical impediments such as:

5.1 Pedagogical Bifurcation: Authenticity vs. Sensation

The greatest danger to IKS integration is the lack of standardized, peer-reviewed, and pedagogically sound content. The curriculum field is currently polarized between two extremes:

- **Entrenched Eurocentric Skepticism:** A significant section of academia, conditioned by decades of Western epistemic training, views any inclusion of ancient Indian thought with deep suspicion, reflexively categorizing it as an unscientific, revivalist project.
- **Chauvinistic Pseudoscientific Claims:** Conversely, enthusiastic but untrained proponents occasionally introduce unverified, hyperbolic assertions into local curricula—such as claiming ancient possession of interplanetary nuclear spacecraft or genetic cloning. This pseudoscience devalues genuine, globally recognized achievements of Indian science (like ancient plastic surgery by Sushruta or linguistics by Panini), exposing the entire framework to academic ridicule.

The lack of rigorous critical-historical methods (Itihasa and Anvikshiki) in filtering text materials presents an ongoing challenge for curriculum designers.

5.2 The Teacher Capacity Deficit

A curriculum is only as good as its teachers. India has an estimated 9.5 million schoolteachers and over 1.5 million higher education faculty. The vast majority have been trained in highly compartmentalized, Westernized paradigms. Most teachers lack the hermeneutic tools to read classical primary texts or understand their historical context. The current Faculty Development Programs (FDPs) run by UGC and AICTE are highly localized and operate at a fractional scale compared to the massive nationwide demand.

5.3 Linguistic Bottlenecks and Translational Loss

The vast majority of primary sources for IKS are written in Sanskrit, Prakrit, Pali, or classical variations of regional languages like Tamil, Telugu, and Persian.

- **Loss of Sanskrit Literacy:** Over generations, India has lost the widespread ability to read and interpret classical Sanskrit in a secular, scientific manner. It is often relegated to ritualistic contexts.
- **Translational Distortion:** When these texts are translated into modern English or vernacular languages, they frequently suffer severe semantic losses. Concepts like Dharma are incorrectly flattened to mean "Religion"; Prakriti is reduced to "Nature"; and Guna is mistranslated as "Attribute." This semantic distortion strips the knowledge system of its underlying logic, making it appear irrational to a modern student.

5.4 Assessment and Evaluation Discordance

Modern educational evaluation is largely driven by standardized multiple-choice questions (MCQs), examinations that assess short-term memory, and narrow parameter-driven grading criteria. However, IKS, by its very nature, emphasizes Pratyaksha (observation), Anumana (inference), Shabda (expert testimony), and deep experiential internalization. Imposing a rigid, Western-style examination format onto a fluid, holistic IKS framework creates a systemic disconnect. Students often end up memorizing the names of ancient scientists and texts to pass exams, completely missing the underlying philosophical insights and critical thinking methodologies.

6. Strategic Opportunities: Transforming Challenges into Global Strengths

If these challenges are met with systematic, data-driven, and institutional solutions, the integration of IKS opens unprecedented opportunities for India's domestic development and global soft power.

6.1 Cognitive Enhancement and Mental Well-being

The global student community is facing an unprecedented crisis of mental health, anxiety, and screen-induced attention deficits. Here, the integration of Yoga Shastra and Darshana (philosophical reflection) offers crucial solutions.

- Integrating mindfulness, Pranayama (breath control), and Yogasanas into daily school curricula provides students with scientifically validated tools for emotional regulation and neurological resilience.
- Furthermore, exposure to Pramana Shastra—the ancient Indian science of debate and logic—can equip modern students to navigate the post-truth era of misinformation by training them to evaluate

claims based on sound epistemic verification.

6.2 Sustainable Development Goals (SDGs) Alignment

The United Nations' SDGs are heavily focused on climate action, responsible consumption, and life on land. Western-led industrial models are increasingly turning to traditional knowledge for sustainable alternatives. A comparison of IKS with SDG is shown in the table:

IKS Domain	Traditional Practice / Philosophy	Modern Sustainability Application	Relevant UN SDG
Environmental Ethics	IsavasyamIdam Sarvam (Sacredness of all matter)	Shift from anthropocentric exploitation to eco-centric co-existence.	SDG 13 (Climate Action) & SDG 15 (Life on Land)
Agriculture	Vrikshayurveda& Organic composting	Reducing chemical fertilizers; implementing indigenous pest control and crop rotation.	SDG 12 (Responsible Consumption & Production)
Water Management	Community-led stepwells (Baolis), Eris, and Ahars	Decentralized water harvesting; combating groundwater depletion without large dams.	SDG 6 (Clean Water & Sanitation)
Architecture	Vastu Vidya & Vernacular materials	Designing bio-climatic, passively cooled buildings using local stone, lime, and mud.	SDG 11 (Sustainable Cities & Communities)

By positioning IKS as a practical blueprint for ecological preservation, India can lead global policy discourse on climate change adaptation.

6.3 Globalizing Indian Higher Education: Study in India

The internationalization of higher education is a key pillar of NEP 2020. However, Indian universities cannot compete with well-funded Western institutions if they merely offer identical, Eurocentric curricula. IKS in this context offers a Unique Selling Proposition (USP). International students are deeply interested in authentic learning tracks in Ayurveda, classical Indian philosophy, computational Sanskrit linguistics, traditional performing arts, and eco-architecture. By creating accredited, high-tech, and rigorous interdisciplinary IKS degrees, Indian universities can attract global scholars, turning the country into an attractive destination for alternative, culturally rich education.

6.4 Computational Linguistics and AI Readiness

Panini's Ashtadhyayi—the ancient Sanskrit grammar system consisting of approximately 4,000 algebraic rules—is widely recognized by computer scientists as the world's first formal, generative language processing engine. As the world enters the era of Artificial Intelligence, Large Language Models (LLMs), and Natural Language Processing (NLP), structural training in Sanskrit linguistics provides a significant

cognitive advantage. Integrating Paninian grammar into modern computer science tracks offers an opportunity to build unique, rule-based computational architectures that could transform NLP and compiler design.

7. Policy Recommendations for Sustainable Implementation

To bridge the gap between policy intent and ground execution, the following strategic interventions are urgently required:

7.1 Establish an Independent National Peer-Review Board

To protect the curriculum from both hyper-skepticism and unverified pseudoscience, an independent, non-partisan National Peer-Review Board for IKS Materials should be created. This board must consist of dual-trained experts—scholars who hold traditional titles (like Mahopadhyaya or Ghanapaathi) alongside modern research credentials (Ph.D.s from recognized global institutions). Every chapter, module, or book introduced into schools must pass rigorous double-blind peer reviews to ensure it is historically accurate and empirically verifiable.

7.2 Launch a National Mission for IKS Teacher Training

The current piecemeal approach to faculty training must be replaced with a scaled, tech-enabled institutional ecosystem.

- **Integrated B.Ed.-IKS Degrees:** The four-year integrated B.Ed. programs mandated by NEP 2020 should include a compulsory specialization track in IKS pedagogy.
- **Digital Micro-Credentials:** A comprehensive library of short, modular, video-based training modules with rigorous assessment criteria should be integrated into the government's DIKSHA platform for existing teachers. Educators should earn salary incentives or promotional credits upon completing these IKS micro-credentials.

7.3 Fund Deep Translation and Cross-Disciplinary Text Projects

The Ministry of Education must launch a heavily funded National Translation Mission dedicated specifically to secular and scientific ancient texts. These translations should preserve the technical meaning of the original terms while providing clear modern equivalents, complete with annotations, digital diagrams, and modern mathematical or chemical proofs.

7.4 Create Flexible, Multi-Modal Assessment Blueprints

The National Assessment Centre, PARAKH, must design new evaluation frameworks that depart from memory-based grading. Assessments for IKS courses should prioritize portfolio-based grading, viva-voce examinations, community field projects, open-book analytical writing, and computational problem-solving. This shift will ensure that students prioritize the conceptual frameworks of IKS over passive memorization.

8. Conclusion: Toward an Epistemically Just Future

The integration of the Indian Knowledge System within the implementation matrix of NEP 2020 is a

historic step toward cultural decolonization and intellectual self-reliance (Atmanirbharta). The initial years of implementation have yielded encouraging successes. However, the path ahead is filled with systemic challenges. Overcoming the teacher capacity deficit, safeguarding the curriculum from unscientific exaggeration, resolving linguistic barriers, and redesigning evaluation frameworks are major tasks that require decades of sustained effort.

Ultimately, the success of IKS does not lie in retreating into the past, but in bringing ancient wisdom to bear on future challenges.

References

- [1]. All India Council for Technical Education. (2022). *Introduction to Indian Knowledge Systems: Concepts and Applications*. PHI Learning.
- [2]. Kapoor, K., & Singh, A. K. (Eds.). (2021). *Indian knowledge systems: Elements, commonality, and security*. Vidya Samiti.
- [3]. Ministry of Education, Government of India. (2020). *National Education Policy 2020*. Ministry of Education.
- [4]. National Council of Educational Research and Training. (2023). *National Curriculum Framework for School Education (NCF-SE)*. NCERT.
- [5]. Raju, C. K. (2007). *Cultural foundations of mathematics: The nature of mathematical proof and the transmission of the calculus from India to Europe in the 16th c. CE*. Pearson Longman.
- [6]. University Grants Commission. (2023). *Guidelines for incorporating Indian Knowledge Systems (IKS) in higher education curricula*. UGC.